



Outline

- Inter-observational Distances
- Partition-based Clustering
- Hierarchical Clustering
- Diagnostics and Optimizing Number of Clusters
- Density-based Clustering



Inter-observational Distances

- The choice of distance metrics is a critical step in clustering
- Pairwise distance calculations can influence the shape of your clusters
- Observations to be clustered can be non-standard (e.g., images, audio signals, etc.), so the process of computing distances first followed by clustering can be an effective approach.



Inter-observational Distances

- Two common distance metrics

$$d_{Euc}(x_i, x_k) = \sqrt{\sum_{j=1}^p (X_{ij} - X_{kj})^2}$$

Eucidean Distance (L_2)

$$d_{Man}(x_i, x_k) = \sum_{j=1}^p |x_{ij} - x_{kj}|$$

Manhattan Distance (L_1)

Inter-observational Distances

- Other (correlation-based) distance metrics

$$d_{Pearson}(x_i, x_k) = 1 - \frac{\sum_{j=1}^p (X_{ij} - \bar{x}_i)(X_{kj} - \bar{x}_k)}{\sqrt{\sum_{j=1}^p (X_{ij} - \bar{x}_i)^2 \sum_{j=1}^p (X_{kj} - \bar{x}_k)^2}}$$

$$d_{Spearman}(x_i, x_k) = 1 - \frac{\sum_{j=1}^p (W_{ij} - \bar{w}_i)(W_{kj} - \bar{w}_k)}{\sqrt{\sum_{j=1}^p (W_{ij} - \bar{w}_i)^2 \sum_{j=1}^p (W_{kj} - \bar{w}_k)^2}}$$

where W_{ij} are the ranks of X_{ij} for feature j , and $\bar{w}_i$ is the average of the ranks for observation i .

*Spearman correlation is used when outliers might be a concern



Distances and Scaling

Consider a data set that looks like the following:

X1	X2	X3	X4
498625	0.53	0.73	1.2
88635	0.12	0.13	2.3
623617	0.43	0.47	2.7
...	...	...	...

The distance between any two observations is basically determined by feature 1. This is an undesired consequence of variables having different scales.

Usually, we want to ensure each variable has equal contribution to the clustering algorithm, and therefore the distance computation.



Distances and Scaling

To address this problem, standardize the variables prior to computing distances

$$\frac{X_{ij} - center(x_j)}{scale(x_j)}$$

where $center(x_j)$ can be the sample mean or median of the j^{th} variable, and $scale(x_j)$ can be the sample standard deviation or mean absolute deviation of the j^{th} variable.



Example: Violent Crimes by State

This data set reports arrests per 100,000 residents for assault, murder, and rape in each of the 50 US states in 1973. Also given is the percentage of population living in urban areas.

	Murder	Assault	UrbanPop	Rape
count	50.000000	50.000000	50.000000	50.000000
mean	7.78800	170.760000	65.540000	21.232000
std	4.35551	83.337661	14.474763	9.366385
min	0.80000	45.000000	32.000000	7.300000
25%	4.07500	109.000000	54.500000	15.075000
50%	7.25000	159.000000	66.000000	20.100000
75%	11.25000	249.000000	77.750000	26.175000
max	17.40000	337.000000	91.000000	46.000000



Example: Violent Crimes by State

First few randomly chosen observations

	State	Murder	Assault	UrbanPop	Rape	StateAbbrev
0	Alabama	13.2	236	58	21.2	AL
1	Alaska	10.0	263	48	44.5	AK
2	Arizona	8.1	294	80	31.0	AZ
3	Arkansas	8.8	190	50	19.5	AR
4	California	9.0	276	91	40.6	CA



Example: Violent Crimes by State

First few observations rescaled (subtract mean, divide by standard deviation)

	Murder	Assault	UrbanPop	Rape
State				
Hawaii	-0.58	-1.51	1.22	-0.11
Indiana	-0.14	-0.70	-0.04	-0.03
New Mexico	0.84	1.38	0.31	1.17
Washington	-0.88	-0.31	0.52	0.54
Maine	-1.32	-1.06	-1.01	-1.45
Alabama	1.26	0.79	-0.53	-0.00



Example: Violent Crimes by State

Euclidean distances calculated on scaled observations

	Hawaii	Indiana	New Mexico	Washington	Maine	Alabama
Hawaii	0.000000	1.561769	3.586656	1.560979	2.743631	3.422932
Indiana	1.561769	0.000000	2.617305	1.152154	2.124266	2.097219
New Mexico	3.586656	2.617305	0.000000	2.504780	4.390177	1.615635
Washington	1.560979	1.152154	2.504780	0.000000	2.655948	2.675068
Maine	2.743631	2.124266	4.390177	2.655948	0.000000	3.520494
Alabama	3.422932	2.097219	1.615635	2.675068	3.520494	0.000000